



PROJECT PROPOSAL

Course Name: Database Management System

Course Code: CSE 224

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Academic Resource Management System

Introduction

Students collect different types of academic materials throughout a semester, such as lecture slides, class notes, assignments, lab reports, and previous question papers. As the amount of material increases, it often becomes difficult to keep everything organized and easy to access when needed. The Academic Resource Management System is a web-based application that aims to provide a centralized platform for managing these resources. Instead of storing materials in scattered folders or locations, users will be able to maintain resource information in a structured database and manage it through a simple interface.

Objectives

The main objectives of this project are:

1. To design and implement a relational database using MySQL.
2. To develop a web-based Academic Resource Management System.
3. To perform full CRUD (Create, Read, Update, Delete) operations.
4. To provide a user-friendly interface for managing academic resources.
5. To connect the frontend, backend, and database into a complete system.
6. To apply data validation and maintain accurate records.
7. To gain practical experience in database management and full-stack development.

Methodology

The project will be developed using Python (Flask), MySQL, HTML, CSS, and JavaScript. The system will be built step by step by designing the database, creating the user interface, implementing CRUD functionalities, and connecting all components together.

Development Process

1. Design the database structure and define relationships between tables.
2. Create tables for resources, subjects, and contributor information.
3. Connect the application with the MySQL database.
4. Implement CRUD operations for managing records.
5. Develop the frontend interface for user interaction.
6. Add validation and test the system functionality.
7. Integrate all components into a complete working application.

System Features

The Academic Resource Management System will include the following features:

- Add new academic resources
- View available resources
- Update existing resource information
- Delete outdated or unwanted resources and organize resources by subject
- Search and filter resources
- Manage contributor information
- Store data in a structured database

Expected Outcome

The expected outcome of this project is a functional Academic Resource Management System that allows users to manage academic materials efficiently through a single platform. The system will make it easier to organize and access study materials while maintaining structured records in a MySQL database. Through this project, we will apply concepts learned in the Database Management System course, including database design, database connectivity, CRUD operations, backend development, and frontend development.

Technologies to be Used

Frontend

- HTML
- CSS
- JavaScript

Backend

- Python (Flask)

Database

- MySQL

Tools

- Visual Studio Code
- XAMPP & MySQL Workbench